

Collecting Data from the Web

Key Data Cleaning Concepts

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What Does “Dirty Data” Actually Mean?

Raw data collected from the web is almost never ready for analysis straight away.

The Spreadsheet Analogy. Imagine a column labelled *Age* that should contain numbers:

Name	Age
Alice	25
Bob	"Unknown"
Charlie	30
Diana	""
Eve	999

- Calculating the **average age** either **crashes** (cannot add “Unknown”) ...
- ... or silently produces a **wrong answer** (because 999 is included as a real value).
- The problem is **invisible** until you inspect the raw values.

Data Types: Why the Computer Thinks “2020” Is Not a Number

To a computer, text and numbers are completely different things — even if they look the same to a human.

Integer (Number)

- Stored as: 2020
- Math works: $2020 + 1 = 2021$
- Comparisons work: $2020 > 2015$

String (Text)

- Stored as: "2020"
- Math fails: $"2020" + "1" = "20201"$
- Comparisons are alphabetical, not numeric

The Key Rule: If even a *single* cell in a column contains text (e.g., "N/A"), the computer converts the *entire column* to text — silently.

The Fix: Detect the column type, remove the offending cells, then convert.

What Is a Missing Value? The Three Disguises

Missing data hides in three different forms. Each must be handled differently.

① The True Missing — NaN / Null

- The computer knows the cell is blank.
- Pandas ignores it safely in calculations (`mean()` skips NaN).
- *Action:* decide whether to drop the row or fill the gap.

② The Empty String — ""

- The computer sees a text value with zero characters. It is **not** blank.
- It is counted as a valid entry and will not be skipped in calculations.
- *Action:* replace "" with NaN first.

③ The Sentinel Value — "N/A", "-99", "Unknown"

- Fake data pretending to be real. Actively corrupts statistics.
- *Action:* identify all sentinel patterns, then replace with NaN.

The Goal: Convert all three disguises into true NaN before any analysis.

Flattening: Turning Nested JSON into a Usable Table

A raw API record is often **nested**: a field contains an entire sub-record inside it. A spreadsheet needs every value in its own column. **Flattening** does that conversion.

Step 1 — Raw JSON record (nested)

```
{
  "id": "W3177828909",
  "title": "Attention Is All You Need",
  "year": 2017,
  "open_access": {
    "is_oa": true,
    "oa_status": "gold"
  },
  "cited_by_count": 120453
}
```

Step 2 —

`pd.json_normalize(records)`

```
pd.json_normalize(records)
```

Step 3 — Flat DataFrame (ready to use)

Column	Value
id	W3177828909
title	Attention Is All You Need
year	2017
open_access.is_oa	True
open_access.oa_status	gold
cited_by_count	120453

The nested `open_access` sub-record becomes two separate columns using **dot**

notation: `open_access.is_oa` and `open_access.oa_status`.

Deduplication: What Is a Duplicate and Why Does It Matter?

When fetching data from multiple pages or sources, the same record often appears more than once.

Exact Duplicates

Every single cell in Row A is identical to Row B.

Solution: Easy — drop the copy instantly with `drop_duplicates()`.

Near Duplicates (The Hard Problem)

Row 1: 10.1038/s41586-021-03819-2

Row 2: <https://doi.org/10.1038/s41586-021-03819-2>

Same paper. Different text. The computer thinks they are unique.

Solution: Standardise the field first (strip the URL prefix), then deduplicate.

Why it matters:

- Duplicates **inflate counts** (50 papers become 70).
- They **distort statistics** (a highly cited paper is counted twice).

What Is a DataFrame? The Spreadsheet Under the Hood

For participants who have never used pandas: a **DataFrame** is simply a spreadsheet held in the computer's memory.

Index	title	year	cited_by_count
0	"Climate change and ..."	2021	142
1	"Large language models ..."	2023	891
2	"Biodiversity loss ..."	2020	57

Like Excel

- Rows = individual records
- Columns = fields / variables
- Index = row number (starts at 0)

Better than Excel

- Handles millions of rows
- Operations run in milliseconds
- Reproducible: every step is code
- Exports to CSV, Excel, JSON, ...

Tidy Data: One Row, One Observation

The **Tidy Data** principle (Hadley Wickham, 2014) defines what a clean table looks like:

- 1 Each **variable** forms one column.
- 2 Each **observation** forms one row.
- 3 Each **cell** contains exactly one value.

Not Tidy

Paper	Years		
Paper A	2020	2021	2022
Paper B	2019	2023	

One cell, multiple values.
Cannot filter or sort by year.

Tidy

Paper	Year
Paper A	2020
Paper A	2021
Paper A	2022
Paper B	2019
Paper B	2023

One row per observation.
Fully filterable and sortable.